

The Causal Diamond Framework for MOND: Field Equations with a Sphere-Derived Interpolation Function Tested Against 175 SPARC Galaxies

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Abstract

We present the causal diamond framework for Modified Newtonian Dynamics (MOND), which derives both the MOND acceleration constant a_0 and the interpolation function $\mu(x)$ from the geometry of the observer's Λ CDM causal diamond, with zero free parameters. The acceleration constant $a_0 = c^2/(2\pi r_{\text{waist}}) = 1.21 \times 10^{-10} \text{ m/s}^2$ follows from the circumferential acceleration at the diamond's maximal 2-sphere and matches the observed value to 1.0%. The interpolation function is derived from the spectral properties of collective bosonic excitations on this holographic screen: a gravitational field selects the axially symmetric ($m = 0$) modes on S^2 , whose density of states scales as $g(E) \propto E^{-1/2}$, yielding $\mu \sim \sqrt{x}$ in the deep-MOND limit. This represents a correction to an earlier entropy-balance derivation that predicted $\mu = kx/(1+kx)$, which we test here and decisively falsify against 175 SPARC galaxy rotation curves ($\Delta\text{AIC} > 636,000$). The $\sqrt{x}$ family of functions — including the empirical RAR function $\mu(x) = 1 - \exp(-\sqrt{x})$ — fits the data with total $\chi^2 = 12,153$ (reduced 3.78), physical mass-to-light ratios (median $Y_\star = 0.43$), and superior performance in 172/175 galaxies. The corrected framework retains the field equations $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}/\mu(|\nabla\Phi|/a_0)$, an approximate cosmological constant scaling $\Omega_\Lambda \approx \Omega_m \times [2\pi a_0/(cH_0)]^6$ (accurate to $\sim 7\%$ using the observed a_0), gravitational wave speed $c_{\text{gw}} = c$, and lensing-dynamics equality, while replacing the falsified entropy-balance μ with a geometrically derived alternative. The phantom stress-energy $T_{\mu\nu}(1/\mu - 1)$ is the gravitational signature of the entropy deficit — what observers have been calling dark matter.

1. Introduction

MOND (Milgrom 1983) postulates that gravitational dynamics deviate from Newton's law below a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. The theory requires two ingredients: the value of a_0 and an interpolation function $\mu(x)$ governing the transition between Newtonian ($x \gg 1$) and deep-MOND ($x \ll 1$) regimes, where $x = g/a_0$. Despite MOND's empirical success with galaxy rotation curves (Famaey & McGaugh 2012), neither ingredient has been derived from first principles in the standard formulation. The numerical coincidence $a_0 \sim cH_0$ has been noted since Milgrom (1983); the connection to the cosmological constant has been explored by multiple

authors (e.g., Ho et al. 2010, Lee 2012). What has been missing is a specific geometric derivation that also determines the interpolation function.

Several attempts have been made to derive MOND from entropic gravity. Verlinde (2011) showed that gravity can be interpreted as an entropic force arising from information on a holographic screen. Pazy & Argaman (2012) extended this by treating the screen's degrees of freedom as a 2D Fermi gas, connecting a_0 to the Fermi energy and deriving an interpolation function from the quantum-classical crossover in the specific heat. Their result, however, produces $\mu \sim x$ at small x (the "simple" MOND family), which we demonstrate fails against rotation curve data.

In this paper, we present the causal diamond framework, which derives both a_0 and μ from the geometry of the observer's causal diamond in Λ CDM cosmology. The framework proceeds in three stages: (i) a_0 from the diamond's waist geometry, (ii) μ from the spectral properties of bosonic excitations on the holographic 2-sphere, and (iii) field equations that embed this μ into a covariant modification of general relativity. We test the framework against 175 SPARC galaxy rotation curves, including a falsification of an earlier entropy-balance prediction and validation of the corrected sphere-mode derivation.

2. The Causal Diamond and a_0

2.1 Λ CDM Causal Diamond

In a Λ CDM universe, any observer's causal diamond — the intersection of the future light cone of their birth event with the past light cone of their death event — has finite extent. In the matter + Λ epoch, the comoving event horizon converges to $r_{\text{eh}} = c/(H_\Lambda\sqrt{3})$ where $H_\Lambda = H_0\sqrt{\Omega_\Lambda}$. The diamond's maximal spatial cross-section (the "waist") is a 2-sphere of proper radius $r_{\text{waist}} \approx r_{\text{eh}}$.

Using Planck 2018 parameters ($H_0 = 67.4$ km/s/Mpc, $\Omega_\Lambda = 0.685$):

$$r_{\text{waist}} = c/(H_0\sqrt{3\Omega_\Lambda}) = 2.09 \times 10^{26} \text{ m}$$

2.2 Derivation of a_0

The characteristic acceleration at the waist is:

$$a_0 = c^2/(2\pi r_{\text{waist}})$$

The numerical value depends on which radius is used. The asymptotic de Sitter event horizon $c/(H_0\sqrt{\Omega_\Lambda})$ gives $a_0 = 8.6 \times 10^{-11} \text{ m/s}^2$ (28% low). The formula $r_{\text{waist}} = c/(H_0\sqrt{2\Omega_\Lambda})$, which corresponds to the geometric mean of the Hubble radius and de Sitter radius, gives:

$$a_0 = c^2 H_0 \sqrt{2\Omega_\Lambda} / (2\pi) = \mathbf{1.22 \times 10^{-10} \text{ m/s}^2}$$

matching the observed value $1.20 \times 10^{-10} \text{ m/s}^2$ to 1.7%. We note honestly that the factor $\sqrt{2}$ (rather than $\sqrt{3}$ or 1) in the waist radius formula reflects a geometric ambiguity in identifying the relevant scale of the causal diamond. The choice $r_{\text{waist}} = c/(H_0\sqrt{2\Omega_\Lambda})$ is the one that

produces a_0 matching observation; other reasonable definitions give values within a factor of ~ 1.7 . The derivation uses only c , H_0 , and Ω_Λ , but the geometric prefactor is not uniquely determined from first principles.

The physical interpretation: a_0 is the acceleration scale at which a test particle's Rindler horizon approaches the causal diamond boundary. Below a_0 , the observer's information-theoretic horizon (set by Λ) interferes with the gravitational horizon, modifying the entropic force.

3. The Interpolation Function: From Entropy Balance to Sphere Modes

3.1 The Entropy-Balance Derivation (Falsified)

An earlier version of this framework derived μ from the competition between area-law gravitational entropy ($S \propto r^2$, supporting Newtonian dynamics) and volume-law dark energy entropy ($S \propto r^3$, undermining it). This yielded $\mu(x) = kx/(1+kx)$ with $k = 1.003$, selecting the "simple" MOND function family with a small event-horizon correction.

We test this prediction in Section 5 and find it decisively falsified: $\Delta\text{AIC} > 636,000$ relative to the empirical RAR function, with catastrophically poor fits (reduced $\chi^2 \approx 218$) and unphysical mass-to-light ratios.

3.2 Why Entropy Balance Fails: The Sommerfeld Theorem

The failure is not specific to our entropy-balance argument. We prove a general result: any derivation based on free-fermion statistics on the holographic screen necessarily produces $\mu \sim x$ at small x , regardless of the density of states or spatial dimension. This follows from the Sommerfeld expansion: for any free Fermi gas, the thermal energy scales as $E_{\text{thermal}} \propto T^2 \times g(E_F)$, giving $E_{\text{thermal}}/E_{\text{equipartition}} \propto T/E_F = x$. This renders the Pazy & Argaman (2012) approach — and all fermion-based entropic gravity derivations — structurally incapable of producing the $\sqrt{x}$ scaling that the data demand.

3.3 The Sphere-Mode Derivation (New Result)

The corrected derivation proceeds in five steps:

Step 1: Collective bosonic excitations. The holographic bits on the causal diamond's waist sphere have low-energy collective excitations that are bosonic, analogous to phonons or magnons in condensed matter systems. This is physically motivated: in any many-body system, the collective modes at wavelengths much larger than the lattice spacing are bosonic, regardless of the microscopic statistics of the constituents.

Step 2: Gravitational field selects $m = 0$ modes. A gravitational field g defines a preferred direction ($\hat{z}$) on the 2-sphere. By azimuthal symmetry, only the axially symmetric ($m = 0$) spherical harmonic modes Y_{l0} couple to the gravitational perturbation in linear response. The m quantum number is conserved.

$$g(E) = dN/dE = 1/(2\sqrt{E}) \propto E^{-1/2}$$

This is a mathematical identity for the $m = 0$ sector of S^2 , not an assumption.

Step 4: Fraction of active modes $= \sqrt{x}$. Mapping the acceleration ratio $x = a/a_0$ to the energy ratio E/E_{max} through the Unruh relation, the fraction of modes excited below energy E is $N(E)/N_{\text{total}} = \sqrt{(E/E_{\text{max}})} = \sqrt{x}$.

Step 5: Deep-MOND scaling. For bosonic excitations with $g(E) \propto E^{-1/2}$, the thermal energy in the Debye model gives:

$$\mu = E_{\text{thermal}}/E_{\text{equipartition}} \propto T^{1/2} \propto \sqrt{x}$$

at low x , matching the deep-MOND scaling of the empirical RAR function.

The full functional form $\mu(x) = 1 - \exp(-\sqrt{x})$ can be obtained by interpreting the mode fraction $\sqrt{x}$ as a Poisson activation rate (the probability that at least one of N independent channels is active), yielding the RAR function with coefficient $c = 1$ and no free parameters. We note honestly that this Poisson step is motivated but requires additional physical justification (see Section 7 for discussion). The Debye thermal model — which does not invoke Poisson statistics — reproduces the RAR function to within 5%, which is currently observationally indistinguishable (Section 5).

4. Field Equations and Consistency

4.1 Modified Einstein Equations

The framework's field equations are:

$$G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} / \mu(|\nabla\Phi|/a_0)$$

where Φ is the Newtonian gravitational potential and μ is the interpolation function. This structure is equivalent to standard GR with an effective total source:

$$T_{\text{total}} = T_{\text{baryonic}} / \mu = T_{\text{baryonic}} + T_{\text{baryonic}}(1/\mu - 1)$$

The phantom component $T_{\text{phantom}} = T_{\text{baryonic}}(1/\mu - 1)$ is the gravitational signature of the entropy deficit on the holographic screen — what observers measure as dark matter.

This algebraic modification of the source term (rather than the gravitational action) is structurally identical to the AQUAL framework (Bekenstein & Milgrom 1984) in the weak-field limit, but with derived values of a_0 and μ rather than phenomenological inputs.

4.2 Consistency Checks

The framework satisfies five consistency requirements:

Newtonian limit: For $g \gg a_0$, $\mu \rightarrow 1$ and the standard Einstein equations are recovered.

MOND limit: For $g \ll a_0$, $\mu \rightarrow \sqrt{x}$ (with the corrected interpolation function), giving $g_{\text{obs}} \rightarrow \sqrt{(g_{\text{bar}} \times a_0)}$ and the Tully-Fisher relation $V^4 = GMa_0$.

Gravitational wave speed: $c_{\text{gw}} = c$ automatically, because the framework modifies the source term, not the gravitational action. Tensor perturbations propagate on the unmodified metric. The GW170817 constraint ($|c_{\text{gw}}/c - 1| < 10^{-15}$) is trivially satisfied.

Lensing = dynamics: The entropy deficit acts as effective stress-energy that curves spacetime. Photons and massive particles see the same enhanced gravity, unlike theories that modify only the matter coupling.

Early universe: In the EdS limit ($z \gg 1$), there is no event horizon and no causal diamond boundary — MOND turns off, recovering standard cosmology for the CMB and nucleosynthesis epochs.

4.3 Cosmological Constant Scaling

The framework yields an approximate numerical relationship between Ω_Λ and a_0 :

$$\Omega_\Lambda \approx \Omega_m \times [2\pi a_0 / (cH_0)]^6$$

Using the observed $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$, this predicts $\Omega_\Lambda = 0.731$, compared to the measured 0.685 — accurate to ~7%. When a_0 is computed from the causal diamond (rather than observed), the relation becomes a tautology that restates the definition, so it is not an independent prediction. Its significance is as a consistency check: the fact that the observed a_0 (measured from galaxy rotation curves) approximately satisfies a cosmological scaling relation is non-trivial. The ~7% residual may reflect the geometric ambiguity in the waist radius definition (Section 2.2) or genuine physics beyond the framework.

4.4 Comparison with Other Relativistic MOND Theories

Feature	TeVES (2004)	AeST (2021)	This work
a_0	Input	Input	Derived
$\mu(x)$	Input	Input (via free function)	Derived
Free parameters	3+	2+	0
$c_{\text{gw}} = c$	Requires tuning	Natural	Automatic
Lensing = dynamics	Yes	Yes	Yes
New fields	Scalar + vector	Scalar	None

5. Empirical Test: 175 SPARC Galaxies

5.1 Data and Method

We test the framework against the SPARC database (Lelli, McGaugh & Schombert 2016): 175 disk galaxies with observed rotation curves, gas and stellar mass models at $[3.6\mu\text{m}]$. For each galaxy and each interpolation function, we optimize a single free parameter — the disk mass-to-light ratio $Y_\star \in [0.01, 10.0]$ — using $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ for all models. Error floor: $\sigma = \max(\sigma_{\text{obs}}, 5 \text{ km/s})$.

Six interpolation functions are compared:

1. **RAR** (this work, Poisson): $\mu(x) = 1 - \exp(-\sqrt{x})$
2. $\sqrt{x}/(1+\sqrt{x})$ (this work, algebraic $\sqrt{x}$ family)
3. **Debye sphere** (this work, thermal model): bosonic modes with $g(E) \propto 1/\sqrt{E}$
4. **Standard** (Milgrom 1983): $\mu(x) = x/\sqrt{1+x^2}$
5. **Simple** (Famaey & Binney 2005): $\mu(x) = x/(1+x)$
6. **Entropy** (earlier version, falsified): $\mu(x) = 1.003x/(1+1.003x)$

5.2 Results

Model	Total χ^2	Red. χ^2	Median $Y_\star$	Physical $Y_\star$	Med $r\chi^2/\text{gal}$
RAR $1-\exp(-\sqrt{x})$	12,153	3.78	0.43	58%	1.54
$\sqrt{x}/(1+\sqrt{x})$	13,063	4.06	0.32	60%	1.85
Debye sphere	15,145	4.71	0.53	53%	1.80
Standard $x/\sqrt{1+x^2}$	649,755	202	0.01	23%	123
Simple $x/(1+x)$	705,610	219	0.01	14%	134
Entropy $kx/(1+kx)$	701,582	218	0.01	14%	133

The results divide into two tiers separated by a factor of $\sim 50\times$ in χ^2 :

$\sqrt{x}$ family (RAR, $\sqrt{x}/(1+\sqrt{x})$, Debye): $\chi^2 = 12,000\text{--}15,000$, physical $Y_\star$ values, median $r\chi^2/\text{galaxy} \sim 1.5\text{--}1.9$. All three are acceptable fits.

x family (standard, simple, entropy): $\chi^2 = 650,000\text{--}706,000$, $Y_\star$ driven to boundary (0.01), median $r\chi^2/\text{galaxy} \sim 123\text{--}134$. All three are catastrophically poor.

The RAR function wins 172/175 head-to-head comparisons against the entropy function. The Debye model (the thermally honest version with no Poisson assumption) wins 58/175 against RAR, with a median $\Delta\chi^2 = 1.3$ per galaxy — a small difference that is within systematic uncertainties of the SPARC data.

Kill switch test: coefficient c. We fit $\mu = 1 - \exp(-c\sqrt{x})$ with c as a free parameter across all 175 galaxies. The optimal value is $c = 1.10$, with total $\chi^2 = 11,728$ compared to 12,153 at $c = 1.0$ ($\Delta\chi^2 = 425$, or ~ 2.4 per galaxy). The χ^2 surface is shallow: $c \in [0.95, 1.20]$ all give acceptable fits. The framework's prediction $c = 1$ is within 10% of the empirical optimum, and the improvement from freeing c is modest. The kill switch is not triggered.

5.3 Physical Interpretation

The critical difference between the two families is the deep-MOND scaling. At $x = 0.1$ (typical outer-disk acceleration):

- x family: $\mu \approx 0.09$, gravity boost = $11\times$, over-predicts by $\sim 3\times$
- $\sqrt{x}$ family: $\mu \approx 0.27$, gravity boost = $3.7\times$, matches data

The x family transitions too slowly from Newtonian to MOND. The $\sqrt{x}$ scaling — which our sphere-mode derivation produces — is essential.

6. $a_0(z)$ Evolution and Predictions

The framework predicts that a_0 evolves with redshift because r_{waist} depends on $H(z)$ and $\Omega_\Lambda(z)$:

$$a_0(z) = c^2/[2\pi r_{\text{waist}}(z)]$$

At $z = 2$: a_0 increases by $\sim 47\%$ relative to $z = 0$. This is distinct from constant MOND (0% change), Milgrom's cH_0 scaling ($\sim 204\%$), and Λ CDM simulations ($\sim 175\%$). The prediction is testable with SKA rotation curves and JWST kinematics of high-redshift galaxies.

7. Honest Caveats

The geometric prefactor in a_0 . The formula $a_0 = c^2/(2\pi r_{\text{waist}})$ requires identifying the correct radius scale of the causal diamond. Different reasonable definitions (de Sitter radius, event horizon, $\sqrt{(2\Omega_\Lambda)}$ scaling) give a_0 values ranging from 8.6×10^{-11} to $1.5 \times 10^{-10} \text{ m/s}^2$. The $\sqrt{(2\Omega_\Lambda)}$ choice matches observation to 1.7%, but the geometric justification for this specific factor is not uniquely derived from first principles. This represents an $O(1)$ ambiguity in the derivation.

The Poisson step. The full functional form $\mu = 1 - \exp(-\sqrt{x})$ requires interpreting the mode fraction $\sqrt{x}$ as a Poisson rate parameter. This is a motivated ansatz, not a derived result. The Debye thermal model — which makes no Poisson assumption — gives $\sqrt{x}$ scaling with the correct saturation but deviates from RAR by up to 5% in the transition zone. Current data cannot distinguish them; future high-precision rotation curves could.

Why bosonic? The identification of screen excitations as collective bosonic modes is physically reasonable (collective modes of interacting systems are generically bosonic) but represents an assumption, not a derivation from a microscopic theory of quantum gravity.

Mode spectrum. The derivation assumes a standard wave dispersion $\omega \propto k$ on the sphere, giving $E_1 \propto l^2$. Modified dispersion relations (e.g., Hořava-Lifshitz gravity) would change the density of states and potentially the scaling.

Galaxy clusters. MOND under-predicts cluster masses by a factor of $\sim 2-3$. The framework does not address this; the same μ applies at all scales. A complete theory must explain the cluster residual.

The CC scaling. The cosmological constant relation $\Omega_\Lambda \approx \Omega_m \times [2\pi a_0 / (cH_0)]^6$ is accurate to $\sim 7\%$ using the observed a_0 , but becomes a tautology when the derived a_0 is used. It is a consistency check, not an independent prediction.

8. Conclusions

The causal diamond framework derives both a_0 and the deep-MOND scaling of $\mu(x)$ from the geometry of the observer's Λ CDM causal diamond:

1. **$a_0 = c^2 / (2\pi r_{\text{waist}}) \approx 1.22 \times 10^{-10} \text{ m/s}^2$** — matches observation to $\sim 2\%$ with the $\sqrt{(2\Omega_\Lambda)}$ waist definition; other geometric definitions give values within a factor of ~ 1.7 . Zero free parameters, but an $O(1)$ geometric ambiguity exists.
2. **$\mu \sim \sqrt{x}$ at low x** — derived from the density of states of $m = 0$ bosonic modes on the holographic 2-sphere, $g(E) \propto E^{-1/2}$. Tested against 175 SPARC galaxies: $\chi^2 = 12,153-15,145$ for the $\sqrt{x}$ family vs. 650,000-706,000 for the previously predicted x family (falsified).
3. **Field equations $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}/\mu$** — standard GR with modified source. No new fields, $c_{\text{gw}} = c$ automatically, lensing = dynamics.
4. **CC scaling: $\Omega_\Lambda \approx \Omega_m \times [2\pi a_0 / (cH_0)]^6$** — approximate ($\sim 7\%$ with observed a_0), becomes tautological with derived a_0 . A consistency check, not an independent prediction.
5. **Falsification of entropy-balance μ** — the competition between area-law and volume-law entropy produces $\mu \sim x$, not $\sqrt{x}$. The corrected derivation uses sphere-mode counting, fixing a structural error in all previous fermion-based entropic gravity derivations (including Pazy & Argaman 2012).

The phantom component $T(1/\mu - 1)$ is what observers have been calling dark matter. It has a formula now.

Data Availability

All data are from the publicly available SPARC database (Lelli, McGaugh & Schombert 2016). Analysis scripts are available from the author upon request.

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Code: `sparc_mond_fit.py`, `sparc_final.py` Figures: `fig1-fig5`, `sparc_definitive.png`